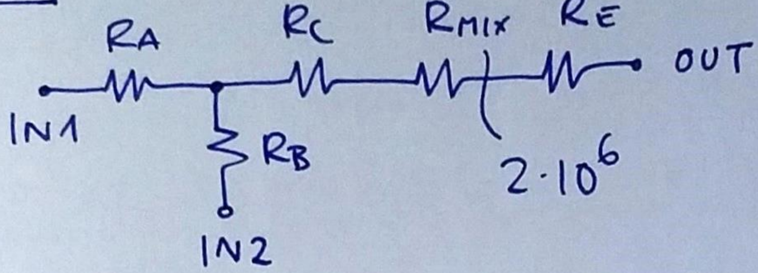
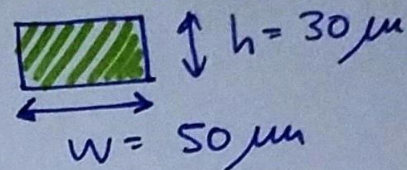
EX. 1

a)





$\frac{w}{h} = \frac{5}{3} = 1.6 > 1.4$

$R = \gamma \cdot L \qquad \gamma = \frac{12 \cdot \eta}{(1-0.63\frac{h}{w}) \cdot h^3 w} = \frac{12 \cdot 10^{-3}}{(1-0.63\frac{3}{5})(30 \cdot 10^{-6})^3 \cdot 50 \cdot 10^{-6}}$

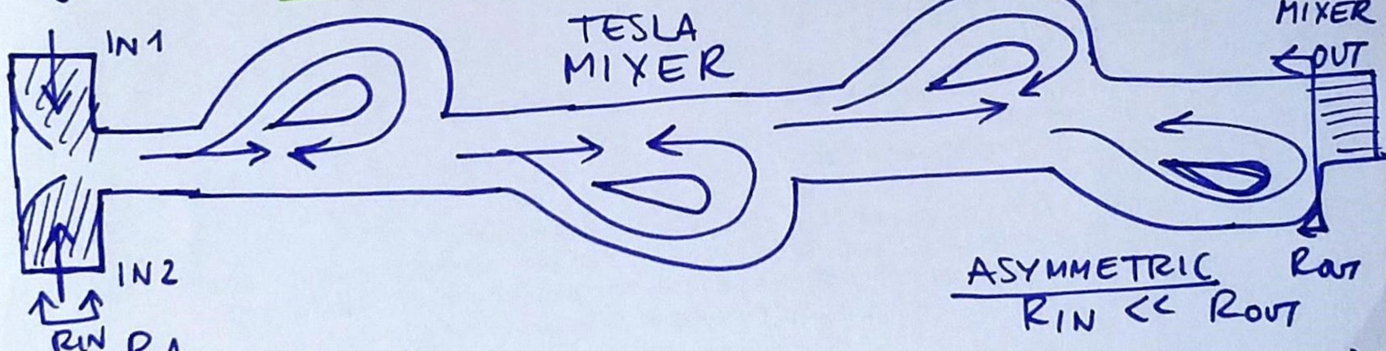
$= 1.4 \cdot 10^{16}$

$R_A = \gamma \cdot A = 2 \cdot 10^{-3} m \cdot 1.4 \cdot 10^{16} = \boxed{2.8 \cdot 10^{13}} \left[\frac{Pa \cdot s}{m^3}\right]$

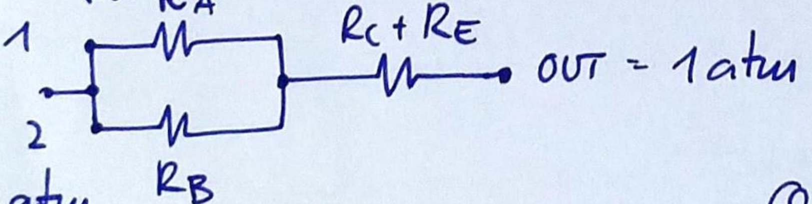
$R_B = R_A/2 = \boxed{1.4 \cdot 10^{13}} = R_C >> R_{MIX} \leftarrow$ NEGLIGIBLE

$R_E = \gamma \cdot 10 = \boxed{14 \cdot 10^{13}}$

b)

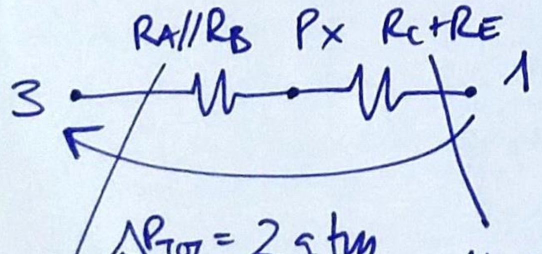


c)



(neglecting MIXER)

$Q_E = \frac{\Delta P_{TOT}}{R_{TOT}} = \frac{2 \cdot 101 \, kPa}{16.3 \cdot 10^{13}}$



$= \boxed{1.24 \cdot 10^{-9} \frac{m^3}{s}}$

$= \boxed{1.24 \frac{\mu l}{s}}$

$= \boxed{74 \, \mu l/min}$

$14 + 1.4 = 15.4 \cdot 10^{13}$

$\frac{2.8 \cdot 1.4}{(2.8 + 1.4)} = 0.93 \cdot 10^{13}$

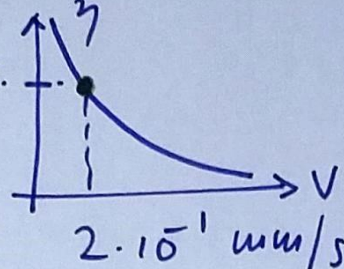
$R_{TOT} = R_A // R_B + R_C + R_E = 16.3 \cdot 10^{13}$

$$P_X = \underbrace{Q_E \cdot (R_C + R_E)}_{1.9 \cdot 10^5} + \underset{\underset{101\,kPa}{\shortparallel}}{P_{OUT}} = \cancel{[illegible]}\ \text{Pa} \quad 291850$$

$$Q_A = \frac{P_{IN} - P_X}{R_A} = \frac{\overset{11\,150}{\cancel{[illegible]}}\ \text{Pa}}{2.8 \cdot 10^{13}} = \boxed{3.98 \cdot 10^{-10}\ m^3/s}$$

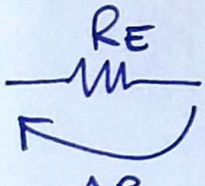
$$+ \qquad = Q_E$$

$$Q_B = \frac{P_{IN} - P_X}{R_B} = 2\,Q_A = \boxed{7{,}96 \cdot 10^{-10}\ m^3/s}$$

d) from plot 

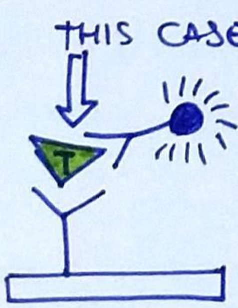
speed $2 \cdot 10^{-1}$ mm/s

$$Q = v \cdot h \cdot w = 2 \cdot 10^{-1} \cdot 10^{-3} \frac{m}{s} \cdot 30 \cdot 10^{-6} m \cdot 50 \cdot 10^{-6} m =$$
$$= 3 \cdot 10^{-13} \frac{m^3}{s}$$

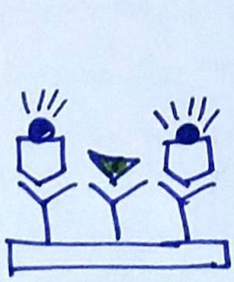


$$\Delta P = R_E \cdot Q = 14 \cdot \cancel{10^{13}} \frac{Pa \cdot \cancel{s}}{\cancel{m^3}} \cdot 3 \cdot \cancel{10^{-13}} \frac{\cancel{m^3}}{\cancel{s}} =$$
$$= \boxed{42\ \text{Pa}} \ \text{(very low)}$$

e) CAPACITANCE ← FLEXIBLE PARTS, COMPRESSIBLE FLUIDS $\left(\frac{\Delta Vol}{\Delta P}\right)$

f) 

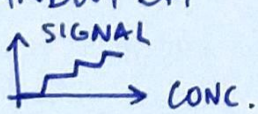
SANDWICH



COMPETITIVE

LABEL FREE → WAVEGUIDE

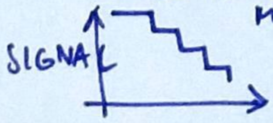
EVANESCENT FIELD

PRO: 

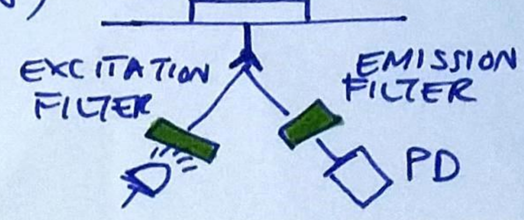
SINGLE SITE

CON: DOUBLE ATTACH SITE for PROBE

NEED EMULATOR MOLECULE

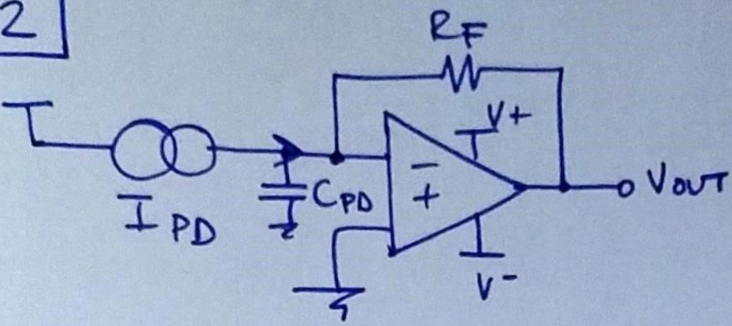


g) FILTERS



$\boxed{\text{EX. 2}}$

a)



SIGN of CURRENT DEPENDS ON BIAS

$$|R_F \cdot I_{max}| = V_{DD} = 5V$$

$$R_F = \frac{5V}{1mA} = \boxed{5k\Omega}$$

b)

$$\sigma = \sqrt{\frac{4kT}{R_F} \cdot \underbrace{\frac{1}{\cancel{2\pi} R_F \cdot C} \cdot \frac{\cancel{\pi}}{2}}_{\text{BW}}} = 400\ \text{pA RMS}$$

NOISE EQ. BW

$\boxed{\text{ADD C IN FEEDBACK TO REDUCE BW}}$

$$\frac{1}{R_F}\sqrt{\frac{\cancel{4}kT}{C \cdot \cancel{4}}} = 400\ pA \qquad \frac{1}{R_F} \cdot \sqrt{\frac{kT}{C}} = 400\ pA$$

$$\frac{kT}{R_F^2} \cdot \frac{1}{(400pA)^2} = C \qquad = \frac{kT}{q} \cdot q \cdot \frac{1}{(R_F \cdot 400pA)^2}$$

$$= \frac{26mV \cdot 1.6 \cdot 10^{-19}}{(5k\Omega \cdot 400pA)^2} = \boxed{1nF} \qquad BW = \frac{1}{2\pi R_F C} = 32kHz$$

c)

$$S_i^2\Big|_{TOT} = i_N^2 + \frac{4kT}{R_F} + \left(\frac{e_n}{R_F}\right)^2 + \left(e_n \cdot 2\pi f \cdot C_{TOT}\right)^2$$

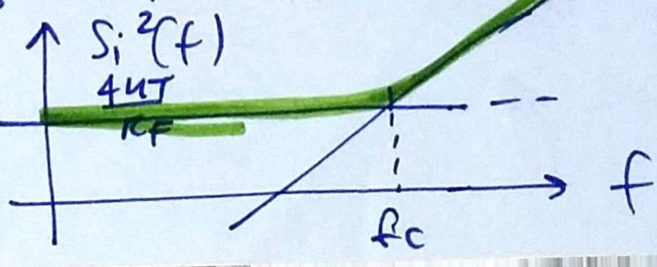
$$C_{TOT} = C_{PD} + C \qquad C_{PD} = 10pF \quad C = 1nF$$

$$\cancel{i_N^2 = \left(\frac{10fA}{\sqrt{Hz}}\right)^2} \quad \text{NEGLIGIBE}$$

$$\frac{4kT}{R_F} = \left(1.8\frac{pA}{\sqrt{Hz}}\right)^2$$

$$\left(\frac{e_n}{5k\Omega}\right)^2 = \left(\frac{400fA}{\sqrt{Hz}}\right)^2$$

$$1.8^2 + 0.4^2 \cdot 1.8^2$$

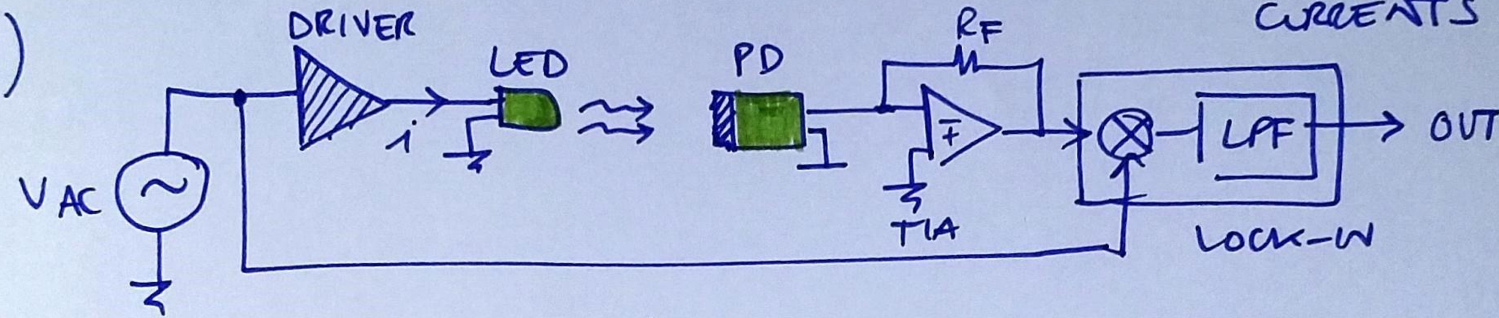


$$\left(1.8\frac{pA}{\sqrt{Hz}}\right)^2 \stackrel{?}{=} \left(\frac{2nV}{\sqrt{Hz}}\cdot 2\pi\cdot f_C\cdot 1nF\right)^2$$

$$f_C = \frac{1.8pA}{2nV\cdot 2\pi\cdot 1nF} = \boxed{143\ kHz}$$

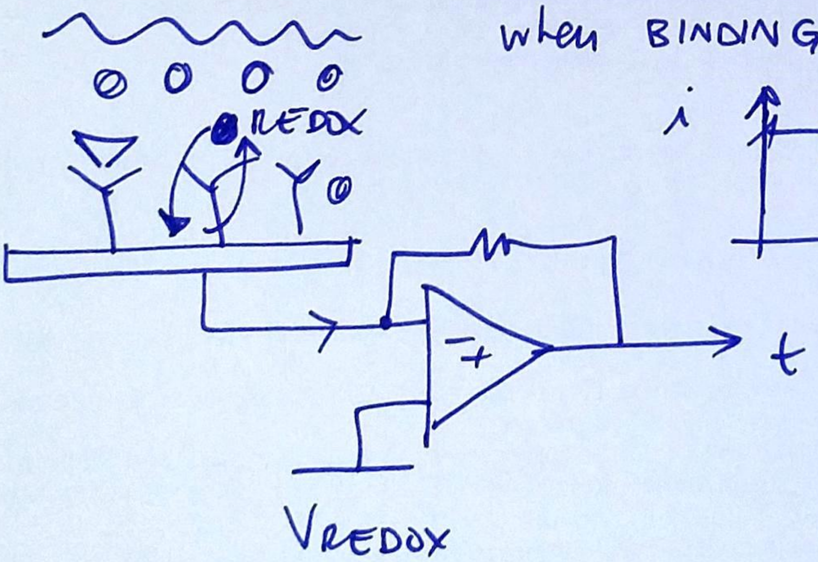
d) $S_i^2 \stackrel{SHOT}{=} 2q\cdot I\big|_{max} = \boxed{\left(17\frac{pA}{\sqrt{Hz}}\right)^2} \gg \frac{4kT}{R_F}$ DOMINANT EFFECT FOR LARGE CURRENTS!

e)

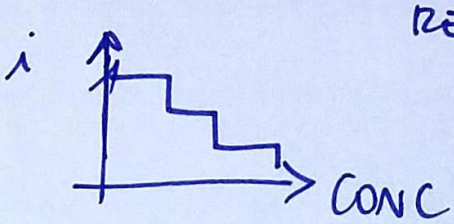


LIGHT MODULATION (OR MECHANICAL SHUTTER)

f) for instace, add a <u>REDOX MOLECULE</u>



when BINDING ⟶ REDOX CURRENT REDUCED



g) TO RELEASE ⟶ <u>INCREASE TEMPERATURE</u> !